

Quiz (Carolina Reaper)

- Q1.** What is the unit rate of 5 meters in 10 seconds? (A)
- (A) $0.5 \frac{m}{s}$
(B) $1 \frac{m}{s}$
(C) $2 \frac{m}{s}$
(D) $5 \frac{m}{s}$
(E) $10 \frac{m}{s}$
- Q2.** A car travels 240 miles in 4 hours. What is its speed in miles per hour? (A)
- (A) $40 \frac{mi}{hr}$
(B) $50 \frac{mi}{hr}$
(C) $60 \frac{mi}{hr}$
(D) $70 \frac{mi}{hr}$
(E) $80 \frac{mi}{hr}$
- Q3.** If y varies directly as x , and $y = 12$ when $x = 4$, what is the constant of variation? (A)
- (A) 2
(B) 3
(C) 4
(D) 6
(E) 8
- Q4.** What is the rate at which a worker can complete a task if $\frac{3}{4}$ of the task is done in 2 hours? (A)
- (A) $\frac{1}{4}$ of task per hour
(B) $\frac{1}{3}$ of task per hour
(C) $\frac{3}{8}$ of task per hour
(D) $\frac{3}{4}$ of task per hour
(E) 1 task per hour
- Q5.** The cost of a pizza is directly proportional to its diameter. If a pizza with a diameter of 16 inches costs 16, what is the cost of a pizza with a diameter of 20 inches? (A)
- 12
- Q6.** A plane flies 480 miles in 3 hours with the wind and 360 miles in 3 hours against the wind. What is the speed of the plane in still air? (A)
- (A) $100 \frac{mi}{hr}$
(B) $120 \frac{mi}{hr}$
(C) $140 \frac{mi}{hr}$
(D) $160 \frac{mi}{hr}$
(E) $180 \frac{mi}{hr}$
- Q7.** A water tank can be filled in 6 hours. How long will it take to fill $\frac{3}{4}$ of the tank at the same rate? (A)
- (A) 2 hours
(B) 3 hours
(C) 4 hours
(D) 4.5 hours
(E) 6 hours
- Q8.** The distance traveled by an object is inversely proportional to its mass. If an object of mass 8 kg travels 12 meters, what distance will an object of mass 6 kg travel? (A)
- (A) 8 meters
(B) 12 meters
(C) 16 meters
(D) 20 meters
(E) 24 meters

Open Ended Questions (FRQ)

- Q9.** A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. The ratio of whole wheat to white bread is 3 : 5. If it takes 2 hours to bake 75 loaves of bread, what is the rate at which they bake whole wheat bread?
- Q10.** A car rental company charges a base fee of 30 plus an additional 0.25 per mile driven. If a customer drives 120 miles, what is their total cost? Show all calculations and explain your reasoning.

Chef's Tips

- Emphasize the importance of reading and following instructions carefully.
- Encourage students to use calculators to check their work when necessary.
- Remind students to show all work and justify their answers for free-response questions